Project Summary: Day 1 OJT – School Data Analysis

1. Dataset Overview

- The dataset contains **5200 rows and 17 columns**.
- It includes information about:
  - School details (ID, District, Block, Urban/Rural)
  - Student performance (% in Math, Language, Science)
  - Teacher-related features (experience, training, ratio)

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2. Data Preprocessing

- Loaded data using **Pandas (read_excel)**.
- Checked:
  - Missing values
  - Data types
  - Basic statistics using .describe()
- Cleaned and prepared data for analysis.

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3. Univariate Analysis

- Analyzed individual variables using:
  - **Histograms**
  - **Bar charts**
  - **Boxplots**
- Key focus:
  - Distribution of student scores
  - Teacher-student ratio variation
  - Teacher experience spread

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4. Bivariate Analysis

- Compared relationships between two variables:
  - Math vs Science scores
  - Teacher experience vs student performance

    - Urban vs Rural performance
- Used:
    - **Scatter plots**
    - **Bar comparisons**
    - **Grouped analysis**

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5. Correlation Analysis

- Calculated correlation matrix to understand relationships.
- Identified:
    - Strong/weak relationships between variables
    - Impact of teacher-related factors on student scores

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6. Key Insights

- Student performance varies across districts and regions.
- Teacher experience shows **positive influence** on student scores.
- Urban schools generally perform better than rural schools.
- Teacher-student ratio impacts performance (lower ratio = better results).
- Some subjects (Math/Science) show moderate correlation.

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7. Tools & Technologies Used

- Python
- Pandas
- Matplotlib / Seaborn
- Jupyter Notebook

Conclusion

This analysis helps in understanding how **teacher quality, school location, and infrastructure** influence student performance. It provides a strong base for **educational decision-making and policy improvements**.